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Privacy-preserving distributed estimation for interconnected dynamic systems ☆

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ABSTRACT

This paper investigates the problem of privacy protection in distributed estimation for interconnected dynamic systems. The exchange of information between subsystems during weighted sum aggregation poses significant privacy risks to distributed estimation. To address these concerns, we propose a noise contamination mechanism for private matrix–vector multiplication and private weighted sum aggregation. This mechanism decomposes the matrix into two column-rank-deficient matrices and injects noises derived from their null spaces to protect the confidentiality of the vector during the matrix–vector multiplication. Several conditions are established to ensure that the mechanism achieves complete confidentiality. Subsequently, we design a double-layer encryption scheme for private weighted sum aggregation with hidden weights, incorporating private matrix–vector multiplication and the Paillier cryptosystem. Furthermore, we develop privacy-preserving distributed estimators for interconnected dynamic systems using the proposed aggregation schemes. Our analysis demonstrates that both subsystems and their distributed estimators achieve effective privacy protection. Finally, an illustrative example is provided to validate the effectiveness of the proposed methods.

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1. Introduction

Cyber–physical systems such as intelligent transportation systems (Li et al., 2017), power networks (Kekatos & Giannakis, 2013), and water distribution networks (Creaco, Campisano, Fontana, Marini, Page, & Walski, 2019) are typically regarded as interconnections of subsystems, which may be either physical entities or purely mathematical artifices. These interconnected dynamic systems (IDSs) form the foundation of human society and exhibit complex system behaviors due to the interactions and couplings among subsystems (Lian, Yang, & Yuan, 2019). It is prohibitive to deploy sensors to measure every state variable in IDSs due to cost constraints and physical limitations. Consequently, accurate estimation of unmeasured states is crucial for the effective synthesis and control of IDSs.

As one of the most important issues in IDSs, distributed state estimation has attracted considerable research interest due to its scalability and low deployment cost (Chen, Hu, Ho, & Yu, 2019, 2022; Farina, Ferrari-Trecate, & Scattolini, 2010; Haber & Verhaegen, 2013; Khan & Moura, 2008; Riverso, Farina, Scattolini, & Ferrari-Trecate, 2013; Riverso, Rubini, & Ferrari-Trecate, 2015; Zhang, Chen, Yu, & Ho, 2022). However, traditional distributed estimation methods face privacy vulnerabilities in two main aspects. First, the distributed communication structure is vulnerable to eavesdropping attacks because subsystem information needs to be exposed to the network environment (Jung, Ryu, & Choi, 2022). Adversaries with limited resources can execute eavesdropping attacks by intercepting and probing information in communication channels, potentially facilitating more sophisticated attacks (Tsiamis, Gatsis, & Pappas, 2020; Yan, Zhang, Xu, & Chen, 2022). For example, it is catastrophic for a multi-robot system to have its location information eavesdropped on by adversaries, especially for military applications. Second, explicit information exchange can breach the privacy of subsystems that wish to keep their information confidential from other honest-but-curious subsystems. For example, user privacy regarding daily activities must be protected from other entities during cooperative state estimation in the smart grid (Giaconi, Gunduz, & Poor, 2018). If one subsystem is compromised, the privacy risk may propagate over the network. Therefore, there is an urgent need to develop privacy-preserving distributed estimators for IDSs.

The most common techniques against eavesdropping include artificial-noise codes using channel state information (Goel & Negi, 2008) and public key cryptography (Shim, 2016). Recent advancements have applied the coding ideas behind these techniques to state estimation problems, such as artificial-noise codes

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using system state information (Tsiamis et al., 2020; Yan et al., 2022). Additionally, cloud-based outsourcing (Darup, Alexandru, Quevedo, & Pappas, 2021) and secure multi-party computation techniques, such as homomorphic encryption (Farokhi, Shames, & Batterham, 2017; Ruan, Gao, & Wang, 2019), algebraic transformation (Sultangazin & Tabuada, 2021), differential privacy (Cortés, Dullerud, Han, Le Ny, Mitra, & Pappas, 2016; Nozari, Tallapragada, & Cortés, 2015, 2017; Ny & Pappas, 2014), secret sharing (Alexandru, Schulze Darup, & Pappas, 2019; Bonawitz et al., 2017; Darup & Jager, 2019; Joye & Libert, 2013; Shamir, 1979; Tjell & Wisniewski, 2020), and artificial-noise obfuscation (Mo & Murray, 2017), are widely employed to safeguard against honest-but-curious entities in systems and control. Private weighted sum aggregation is a crucial problem in distributed estimation that is closely related to secure multi-party computation. In this problem, agent parties send their contributions to an aggregating party to compute a weighted sum of contributions (Alexandru & Pappas, 2022). Most existing literature focuses on the situation where the weights are known only to the agent parties (Bonawitz et al., 2017; Joye & Libert, 2013; Tjell & Wisniewski, 2020). For example, a private weighted sum aggregation scheme was introduced in Joye and Libert (2013), combining secret sharing with the Paillier cryptosystem. Another scenario occurs when the weights are known only to the aggregating party, which can be addressed by combining secret sharing and functional encryption for inner products (Agrawal, Libert, & Stehlé, 2016). However, these schemes fail on a single agent due to the limitations of secret sharing and are inefficient in communication when adding or removing agents because of the need to regenerate secret shares. The most challenging scenario is when neither the agent parties nor the aggregating party knows the weights. Double-layer encryption methods have been proposed in (Alexandru & Pappas, 2022; Alexandru et al., 2019; Darup, Redder, & Quevedo, 2019) to address this issue, though they inherit the drawbacks of secret sharing and require additional encryption operations for resource-constrained agents. Furthermore, exposing the private key used for encrypting hidden weights to the aggregating party poses significant risks.

Motivated by the analysis above, we investigate the problem of privacy-preserving distributed estimation for IDSs from the perspective of private weighted sum aggregation. The contributions of this paper are summarized as follows.

- **Private Matrix–vector Multiplication.** We propose a noise contamination mechanism to achieve private matrix–vector multiplication. This mechanism decomposes the matrix into two column-rank-deficient matrices for generating injection noises to protect the privacy of the vector during the matrix–vector multiplication. We then specify conditions that ensure the multiplication process achieves the desired level of privacy, known as complete confidentiality. These conditions can always be satisfied through appropriate matrix decomposition. Compared with the artificial noise method in Goel and Negi (2008), the proposed noise contamination mechanism offers a higher degree of privacy protection because it guarantees that every component of the vector is protected under the complete confidentiality conditions. Moreover, our mechanism is applicable to general matrix–vector multiplications, not limited to cases where the matrix is rank-deficient.
- **Private Weighted Sum Aggregation.** We design two private weighted sum aggregation schemes, one with centralized weights and one with hidden weights. Compared to existing schemes, our proposals avoid the drawbacks of secret sharing under a single agent or variable structures. Additionally, the scheme with hidden weights is computationally efficient for agents as it does not involve encryption operations.
- **Privacy-Preserving Distributed Estimator.** We develop two privacy-preserving distributed estimators using private weighted sum aggregation schemes. It is demonstrated that subsystems and their distributed estimators with third-party outsourcing are effectively privacy-protected.

Notation: Throughout this paper, we use the following notations. $\mathbb{R}$ is the set of real numbers, and $\mathbb{Z}_+$ is the set of positive integers. Define $\mathbb{Z}_n := \{0, 1, \ldots, n-1\}$ and $\mathbb{N}_L := \{1, 2, \ldots, L\}$. The identity matrix is $\mathbf{I}$, and the zero matrix is $\mathbf{0}$. The notation $A_{i,:}$ ($A_{:,i}$) is used to represent the ith row (column) of a matrix A. The transpose, rank, null space, and span space of a matrix A are represented by A^{T}, $\mathrm{rank}(A)$, $\mathrm{null}(A)$, and $\mathrm{span}(A)$, respectively. The orthogonal complement of a subspace S is denoted as $S^{\perp}$, while the notation $X \succ 0$ means that X is a positive definite matrix. Moreover, the infinite norm of a vector x is represented as $\|x\|_\infty$. Given $a, b \in \mathbb{Z}_+$, $a \mod b$ denotes the remainder of integer division, while $\gcd(a, b)$ is the greatest common divisor of a and b.

2. Preliminaries

Let us first introduce the Paillier cryptosystem. The Paillier cryptosystem is a public-key cryptosystem based on the intractable composite residuosity class problem. A simplified version of the Paillier cryptosystem is described as follows (Ruan et al., 2019):

- Key generation
  - Randomly and independently select two large prime numbers $p_1 \in \mathbb{Z}_+$ and $p_2 \in \mathbb{Z}_+$ with the same bit length;
  - Compute the public key $k_p = p := p_1 p_2$;
  - Determine the private key $k_s = (\lambda, \mu)$, where $\lambda = (p_1 - 1)(p_2 - 1)$, and μ is the modular multiplicative inverse of λ, given by $\mu = \lambda^{-1} \mod p$.
- Encryption $c = \mathcal{E}(m) \in \mathbb{Z}_{p^2}$
  - Randomly select $r \in \mathbb{Z}_p^* := \{x \in \mathbb{Z}_p \mid \gcd(x, p) = 1\}$;
  - Calculate the ciphertext of a message $m \in \mathbb{Z}_p$ by $\mathcal{E}(m) = ((p+1)^m r^p) \mod p^2$.
- Decryption $m = \mathcal{D}(c) \in \mathbb{Z}_p$
  - Calculate the plaintext of any ciphertext $c \in \mathbb{Z}_{p^2}$ by $\mathcal{D}(c) = \left(\mathcal{L}(c^\lambda \mod p^2)\mu\right) \mod p$, where $\mathcal{L}(x) = (x-1)/p$ is an integer division function.

In the Paillier cryptosystem, the public key k_p is shared by all parties for encrypting messages, but the private key k_s is accessible only to trusted parties for decrypting messages. The Paillier cryptosystem can provide semantic security, also known as indistinguishability under chosen plaintext attacks. Semantic security implies that any information revealed cannot be feasibly extracted, making it a fundamental requirement for encryption systems. Additionally, the Paillier cryptosystem is additively homomorphic, meaning that addition operations on plaintexts can be performed directly on the ciphertexts. The additive homomorphism follows from

$$\begin{aligned}\mathcal{E}(m_1)\mathcal{E}(m_2) &= \left((p+1)^{m_1} r_1^p (p+1)^{m_2} r_2^p\right) \mod p^2 \\ &= \left((p+1)^{m_1+m_2}(r_1 r_2)^p\right) \mod p^2 \\ &= \mathcal{E}(m_1 + m_2).\end{aligned} \tag{1}$$

Notice that a multiplication operation can be represented as finite addition operations, implying that

$$\mathcal{E}(m_1)^{m_2} = \mathcal{E}(m_1 m_2). \tag{2}$$

Given a matrix M, element-wise encryption and decryption are respectively denoted by $\mathcal{E}(M)$ and $\mathcal{D}(M)$, where $[\mathcal{E}(M)]_{i,j} := \mathcal{E}(M_{i,j})$ and $[\mathcal{D}(M)]_{i,j} := \mathcal{D}(M_{i,j})$. Given matrices $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times n}$, and $C \in \mathbb{R}^{n\times r}$, the properties (1) and (2) can be used to define operations $\mathcal{E}(A) \boxplus \mathcal{E}(B)$ and $\mathcal{E}(A) \boxtimes C$ such that:

$$\begin{aligned}[\mathcal{E}(A) \boxplus \mathcal{E}(B)]_{i,j} : &= [\mathcal{E}(A)]_{i,j}\,[\mathcal{E}(B)]_{i,j} \\ &= \mathcal{E}(A_{i,j})\mathcal{E}(B_{i,j}) \\ &= \mathcal{E}([A+B]_{i,j}) = [\mathcal{E}(A+B)]_{i,j}\end{aligned} \tag{3a}$$

$$\begin{aligned}[\mathcal{E}(A) \boxtimes C]_{i,j} : &= \prod_{q=1}^{n} \left([\mathcal{E}(A)]_{i,q}\right)^{C_{q,j}} \\ &= \mathcal{E}\left(\sum_{q=1}^{n} A_{i,q}C_{q,j}\right) \\ &= \mathcal{E}\left([AC]_{i,j}\right) = [\mathcal{E}(AC)]_{i,j}\,.\end{aligned} \tag{3b}$$

It follows from these definitions that $\mathcal{E}(A) \boxplus \mathcal{E}(B) = \mathcal{E}(A+B)$ and $\mathcal{E}(A) \boxtimes C = \mathcal{E}(AC)$.

Then, the concept of a zonotope is recalled. An r-order zonotope $\mathcal{Z} \subset \mathbb{R}^n$ is defined as follows:

$$\mathcal{Z} := \{z \in \mathbb{R}^n \mid z = c + Ru,\ \ \|u\|_\infty \le 1\} = \langle c, R\rangle \tag{4}$$

where $c \in \mathbb{R}^n$ and $R \in \mathbb{R}^{n\times r}$ are the center and the generator matrix of $\mathcal{Z}$, respectively. The notation $\langle\cdot,\cdot\rangle$ is a two-tuple used to represent a zonotope. Note that the zonotope $\mathcal{Z}$ is an affine transformation of a unit hypercube $\mathcal{B}_r := \langle \mathbf{0}, \mathbf{I}_r\rangle$.

3. Problem formulation

Consider a discrete-time interconnected system $\mathbf{S}$ composed of L subsystems, where the state and measurement dynamics of the ith subsystem are described by

$$\mathbf{S}_i : \begin{cases} x_i(k+1) = A_i(k)x_i(k) + \sum\limits_{i_\rho \in \Omega_i} A_{i,i_\rho}(k)x_{i_\rho}(k) \\ \qquad\qquad + B_i(k)w_i(k) \\ y_i(k) = C_i(k)x_i(k) + D_i(k)v_i(k) \end{cases} \tag{5}$$

where $x_i(k) \in \mathbb{R}^{n_i}$, $y_i(k) \in \mathbb{R}^{m_i}$, $w_i(k) \in \mathbb{R}^{n_{w_i}}$, and $v_i(k) \in \mathbb{R}^{n_{v_i}}$ are the state, the measurement, the process noise, and the measurement noise of the ith subsystem $\mathbf{S}_i$, respectively. The system matrices $A_i(k)$, $A_{i,i_\rho}(k)$, $B_i(k)$, $C_i(k)$, and $D_i(k)$ are with appropriate dimensions. Let us define $\hat{D}_i(k) := D_i(k)D_i^{\mathrm{T}}(k)$ and assume that $\hat{D}_i(k) \succ 0$. The neighboring subsystems of $\mathbf{S}_i$ are represented as $\mathbf{S}_{i_1}, \ldots, \mathbf{S}_{i_\rho}, \ldots, \mathbf{S}_{i_{\theta_i}}$ with $\theta_i < L$ being the number of neighbors, and their indices are characterized by the set Ω_i. The following assumptions regarding communication, noises, and initial states are provided.

Assumption 1. Each subsystem can communicate with its neighbors.

Assumption 2. The initial state $x_i(0)$ and the noises $w_i(k)$ and $v_i(k)$ are wrapped within zonotopic sets:

$$\begin{aligned} &w_i(k) \in \mathcal{B}_{n_{w_i}} \\ &v_i(k) \in \mathcal{B}_{n_{v_i}} \\ &x_i(0) \in \langle c_{i,0}, R_{i,0}\rangle \subset \mathbb{R}^{n_i}. \end{aligned} \tag{6}$$

Zonotopic estimators are sufficiently simple to be implemented in distributed systems and can avoid the wrapping effect (Combastel, 2015). Therefore, we focus on the problem of privacy protection in a distributed zonotopic estimator. However, the results can also be extended to other distributed estimators. The distributed estimation problem for (5) is to design the following recursive estimator to estimate the real states of this interconnected system:

$$\hat{\mathbf{S}}_i : \begin{cases} \hat{\mathcal{X}}_i^p(k) = f_i\left(\hat{\mathcal{X}}_i(k-1), \hat{\mathcal{X}}_{i_1}(k-1), \ldots, \hat{\mathcal{X}}_{i_{\theta_i}}(k-1)\right) \\ \hat{\mathcal{X}}_i(k) = g\left(\hat{\mathcal{X}}_i^p(k), y_i(k)\right) \end{cases} \tag{7}$$

where $\hat{\mathcal{X}}_i^p(k) := \langle \hat{x}_i^p(k), R_i^p(k)\rangle$ and $\hat{\mathcal{X}}_i(k) := \langle \hat{x}_i(k), R_i(k)\rangle$ are the prediction and the observation update (local estimate) of the local state $x_i(k)$, respectively. The functions $f_i(\cdot)$ and $g(\cdot)$ represent prediction step and observation update step, respectively. Note that the private data in subsystem $\mathbf{S}_{i_\rho}$ ($i_\rho \in \Omega_i$) is exposed to subsystem $\mathbf{S}_i$ during the distributed estimation process described in (7). Concretely, the prediction step $f_i(\cdot)$ requires information from neighboring subsystems and will lead to data leakage problems during computation, while the observation update step $g(\cdot)$ involves only local information processing and does not suffer such privacy issues. In this paper, we focus on the following distributed zonotopic estimator in plaintext (Zhang, Chen, & Yu, 2023), which includes the prediction step and the observation update step:

Prediction: By the process equation, the prediction for $x_i(k)$ is calculated by

$$\begin{aligned} \hat{x}_i^p(k) =& A_i(k-1)\hat{x}_i(k-1) \\ &+ \sum_{i_\rho \in \Omega_i} A_{i,i_\rho}(k-1)\hat{x}_{i_\rho}(k-1) \\ R_i^p(k) =& \big[A_i(k-1)R_i(k-1) \ \ A_{i,i_1}(k-1)R_{i_1}(k-1) \\ &\ldots \ \ A_{i,i_{\theta_i}}(k-1)R_{i_{\theta_i}}(k-1) \ \ B_i(k-1)\ \big]\,. \end{aligned} \tag{8}$$

Observation update: When the measurement $y_i(k)$ is observed, the observation update for $x_i(k)$ can be obtained as

$$\begin{aligned} \hat{x}_i(k) &= \hat{x}_i^p(k) + K_i(k)\left[y_i(k) - C_i(k)\hat{x}_i^p(k)\right] \\ R_i(k) &= \left[(\mathbf{I} - K_i(k)C_i(k))R_i^p(k) \quad -K_i(k)D_i(k)\right]. \end{aligned} \tag{9}$$

Here, $K_i(k)$ is a gain matrix used to turn the uncertainty propagation. It provides flexibility to balance the influence of the system noise $w_i(k)$ and the measurement noise $v_i(k)$. Under the performance index $J(R_i(k)) := \|R_i(k)\|_{F,W}^2$ with $W \succ 0$ and $\|\cdot\|_{F,W}$ being the weighted Frobenius norm (Combastel, 2015), an optimal gain matrix $K_i^{\mathrm{opt}}(k)$ can be calculated as follows:

$$\begin{aligned} K_i^{\mathrm{opt}}(k) &= R_i^p(k)\left[R_i^p(k)\right]^{\mathrm{T}} C_i^{\mathrm{T}}(k) \\ &\times \left\{C_i(k)R_i^p(k)\left[R_i^p(k)\right]^{\mathrm{T}} C_i^{\mathrm{T}}(k) + \hat{D}_i(k)\right\}^{-1}. \end{aligned} \tag{10}$$

The stability analysis for the distributed zonotopic estimator was conducted in Zhang et al. (2023).

For the distributed zonotopic estimator, the prediction step $f_i(\cdot)$ contains a weighted sum aggregation term $\sum_{i_\rho \in \Omega_i} A_{i,i_\rho}(k-1)\hat{x}_{i_\rho}(k-1)$. As discussed, this term is the leading cause of privacy problems in distributed estimation for IDSs. The neighboring subsystems may not wish to disclose their local estimates $\hat{x}_{i_\rho}(k-1)$ ($i_\rho \in \Omega_i$) to subsystem $\mathbf{S}_i$ during the aggregation. Therefore, additional privacy protection schemes are necessary for (8)–(9) to design privacy-preserving distributed estimators. To be more specific about privacy issues, the following rigorous definition of estimator privacy security is used in the context.

Definition 1. The distributed estimator $\hat{\mathbf{S}}_i$ is privacy-protected if its private estimates $\hat{\mathcal{X}}_i^p(k)$ and $\hat{\mathcal{X}}_i(k)$ are unavailable to other

subsystems or untrusted entities during the prediction step and the observation update step.

To achieve the privacy goal in Definition 1, the problem of private weighted sum aggregation should be addressed first. It is also important to note that private weighted sum aggregation is closely related to private matrix–vector multiplication, which is essential for protecting the privacy of $\hat{x}_{i_\rho}(k-1)$ during the computation $A_{i,i_\rho}(k-1)\hat{x}_{i_\rho}(k-1)$.

Consequently, the objectives of this paper are described as follows.

(1) Design a private matrix–vector multiplication mechanism for private weighted sum aggregation such that no private information other than the aggregation results is exposed.
(2) Develop privacy-preserving distributed estimators for the interconnected system (5) using the proposed private weighted sum aggregation schemes. The designed distributed estimators should meet the privacy requirements specified in Definition 1.

4. Main results

In this section, we first introduce a noise contamination mechanism for private matrix–vector multiplication. Subsequently, we apply this mechanism to design private weighted sum aggregation schemes and develop privacy-preserving distributed estimators for IDSs.

4.1. Private matrix–vector multiplication for weighted sum aggregation

Let us begin by examining the issues related to private matrix–vector multiplication and private weighted sum aggregation.

Private Matrix–vector Multiplication. Consider a system consisting of one agent and one client. The agent possesses data $x \in \mathbb{R}^n$, while the client aims to compute the matrix–vector multiplication $x_m = Mx$ using a known matrix $M \in \mathbb{R}^{m\times n}$. The privacy requirement for private matrix–vector multiplication is as follows: the client should have access only to the computed result x_m and not be able to infer the agent's data x.

Private Weighted Sum Aggregation (Alexandru & Pappas, 2022). Consider a system consisting of L agents and one aggregator. Each agent possesses data $x_i \in \mathbb{R}^n$ ($i \in \mathbb{N}_L$), and the aggregator aims to compute an aggregate $x_a \in \mathbb{R}^m$ defined by $x_a = \sum_{i=1}^{L} M_i x_i$, where $M_i \in \mathbb{R}^{m\times n}$. The privacy requirement for private weighted sum aggregation is as follows: the aggregator should have access only to the aggregation result x_a and not be able to infer the agents' data x_i ($i = 1, 2, \ldots, L$).

The problem of private weighted sum aggregation varies depending on whether the weights M_i ($i = 1, 2, \ldots, L$) are known only to the agents, only to the aggregator (centralized weights), or to neither (hidden weights). Private weighted sum aggregation with hidden weights is more challenging because both the weights M_i and the data x_i need to be encoded for transmission. Furthermore, the computation of the weighted sum aggregation should keep confidential from the aggregator, which can only be achieved using homomorphic encryption algorithms, such as the Paillier cryptosystem. The following lemma demonstrates that third-party outsourcing can transform the problem of private weighted sum aggregation with centralized weights into the problem with hidden weights.

Lemma 1. *In private weighted sum aggregation with centralized weights, if the aggregator outsources the computation $x_a = \sum_{i=1}^{L} M_i x_i$ to a third-party while keeping the weights confidential, then the third-party is tasked with solving a private weighted sum aggregation problem with hidden weights.*

In what follows, we will propose a noise contamination mechanism to address the problem of private matrix–vector multiplication. This mechanism relies on matrix decomposition and noise injection techniques. We begin with the following lemma.

Lemma 2. *Given a non-zero matrix $M \in \mathbb{R}^{m\times n}$ ($n > 1$), it can be decomposed into the sum of two matrices, both of which are column-rank-deficient (i.e., $\mathrm{rank}(M) < n$).*

Proof. Let $\mathrm{rank}(M) = r \le n$. All row vectors of M can be represented as linear combinations of π ($r \le \pi \le n, \pi > 1$) linearly independent basis vectors $b_j \in \mathbb{R}^n$ ($j = 1, 2, \ldots, \pi$), *i.e.*,

$$\left[M_{i,:}\right]^{\mathrm{T}} = \sum_{j=1}^{\pi} \alpha_{i,j} b_j. \tag{11}$$

Hence, the matrix M can be represented as follows:

$$M = \underbrace{\begin{bmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1\pi} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2\pi} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{m\pi} \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} b_1^{\mathrm{T}} \\ b_2^{\mathrm{T}} \\ \vdots \\ b_\pi^{\mathrm{T}} \end{bmatrix} = \sum_{j=1}^{\pi} \mathcal{A}_{:,j} b_j^{\mathrm{T}}. \tag{12}$$

Then the matrix M can be decomposed into the sum of two matrices $M = M_a + M_b$ ($M_a, M_b \in \mathbb{R}^{m\times n}$) with

$$M_a = \mathcal{A}_{:,1} b_1^{\mathrm{T}}, \quad M_b = \sum_{j=2}^{\pi} \mathcal{A}_{:,j} b_j^{\mathrm{T}}. \tag{13}$$

This decomposition requires that $\mathcal{A}_{:,1}$ is a non-zero vector, and at least one of $\mathcal{A}_{:,j}$ ($j = 2, \ldots, \pi$) is a non-zero vector. It is also known from (12) that

$$M^{\mathrm{T}} = \begin{bmatrix} b_1 & b_2 & \cdots & b_\pi \end{bmatrix} \begin{bmatrix} \mathcal{A}_{:,1}^{\mathrm{T}} \\ \mathcal{A}_{:,2}^{\mathrm{T}} \\ \vdots \\ \mathcal{A}_{:,\pi}^{\mathrm{T}} \end{bmatrix}. \tag{14}$$

If the equation $\begin{bmatrix} b_1 & \cdots & b_{j-1} & b_{j+1} & \cdots & b_\pi \end{bmatrix} X = M^{\mathrm{T}}$ has no solution, *i.e.*,

$$\begin{aligned} &\mathrm{rank}\left(\begin{bmatrix} b_1 & \cdots & b_{j-1} & b_{j+1} & \cdots & b_\pi & M^T \end{bmatrix}\right) \\ &\qquad \neq \mathrm{rank}\left(\begin{bmatrix} b_1 & \cdots & b_{j-1} & b_{j+1} & \cdots & b_\pi \end{bmatrix}\right), \end{aligned} \tag{15}$$

then $\mathcal{A}_{:,j}^{\mathrm{T}}$ ($j \in \mathbb{N}_\pi$) must be a non-zero vector. When $\pi = r \ge 2$, all vectors $\mathcal{A}_{:,j}$ ($j = 1, \ldots, \pi$) are non-zero vectors. When $\pi > r \ge 2$, the basis vector b_1 should be carefully selected according to (15) by reordering b_j ($j = 1, 2, \ldots, \pi$). When $\pi > r = 1$, all basis vectors cannot be multiples of $\left[M_{i,:}\right]^{\mathrm{T}}$, and the basis vector b_1 also needs to be selected by reordering b_j ($j = 1, 2, \ldots, \pi$). Moreover, since b_j ($j = 1, 2, \ldots, \pi$) are linearly independent basis vectors, it follows that $\mathrm{rank}(M_a) = 1 < n$ and $\mathrm{rank}(M_b) \le n - 1 < n$, which indicates M_a and M_b are column-rank-deficient. ◇

Example 1. Consider the matrix $M = \begin{bmatrix} 1 & 0 \end{bmatrix}$ with $\mathrm{rank}(M) = 1$. If the basis vectors are selected as $b_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^{\mathrm{T}}$ and $b_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}^{\mathrm{T}}$, then M cannot be decomposed as described in Lemma 2 because b_1 is equal to $\left[M_{1,:}\right]^{\mathrm{T}}$. However, if the basis vectors are selected as $b_1 = \begin{bmatrix} 1 & 1 \end{bmatrix}^{\mathrm{T}}$ and $b_2 = \begin{bmatrix} 1 & -1 \end{bmatrix}^{\mathrm{T}}$, then M can be decomposed as $M = 0.5 b_1^{\mathrm{T}} + 0.5 b_2^{\mathrm{T}}$.

The decomposed column-rank-deficient matrices have the property that the vectors constructed from their null space have non-zero elements. This property can be used for private matrix–vector multiplication. The following lemma presents a noise contamination mechanism.

Lemma 3. *Given a non-zero matrix $M \in \mathbb{R}^{m\times n}$ and a non-zero vector $x \in \mathbb{R}^n$, the private matrix–vector multiplication Mx can be achieved using the following noise contamination mechanism:*

(1) If $n > 1$: Compute $M = M_a + M_b$ by Lemma 2. The private matrix–vector multiplication Mx is achieved by $M_a(x+N_a\mathbf{n}_a)+M_b(x+N_b\mathbf{n}_b)$, where $\text{span}(N_a) = \text{null}(M_a)$, $\text{span}(N_b) = \text{null}(M_b)$, and randomly generated noises $\mathbf{n}_a$ and $\mathbf{n}_b$ are with appropriate dimensions. During the computation, the vector x is contaminated by noise terms $N_a\mathbf{n}_a$ and $N_b\mathbf{n}_b$ to keep its privacy, resulting in $x + N_a\mathbf{n}_a$ and $x + N_b\mathbf{n}_b$.

(2) If $n = 1$ and $m > 1$: Compute $M^{\mathrm{T}} = M_a + M_b$ by Lemma 2. The private matrix–vector multiplication Mx is achieved by $(M_a(x\mathbf{I}_m + N_a\mathbf{n}_a) + M_b(x\mathbf{I}_m + N_b\mathbf{n}_b))^{\mathrm{T}}$, where x is concealed into $x\mathbf{I}_m + N_a\mathbf{n}_a$ and $x\mathbf{I}_m + N_b\mathbf{n}_b$.

(3) If $n = m = 1$: Augment M with a zero column vector and compute $\begin{bmatrix} M & \mathbf{0}\end{bmatrix} = M_a + M_b$ by Lemma 2. The private matrix–vector multiplication of Mx is achieved by $M_a(x\mathbf{I}_2 + N_a\mathbf{n}_a) + M_b(x\mathbf{I}_2 + N_b\mathbf{n}_b)$, where x is concealed into $x\mathbf{I}_2 + N_a\mathbf{n}_a$ and $x\mathbf{I}_2 + N_b\mathbf{n}_b$.

Proof. In the first case, it is known from Lemma 2 that M_a and M_b are column-rank-deficient, implying $\text{null}(M_a) \neq \{\mathbf{0}\}$ and $\text{null}(M_b) \neq \{\mathbf{0}\}$. This means that $N_a\mathbf{n}_a \neq \mathbf{0}$ and $N_b\mathbf{n}_b \neq \mathbf{0}$ are effective noise injections. By the properties of the null space, one has that $M_aN_a\mathbf{n}_a = \mathbf{0}$ and $M_bN_b\mathbf{n}_b = \mathbf{0}$. Thus, the equation $Mx = M_a(x+N_a\mathbf{n}_a)+M_b(x+N_b\mathbf{n}_b)$ holds. In the second case, Mx can be represented as $\left(M^{\mathrm{T}}x\mathbf{I}_m\right)^{\mathrm{T}}$ because x is a scalar. In the third case, one can augment M with a zero column vector, yielding $Mx = \begin{bmatrix} M & \mathbf{0}\end{bmatrix}x\mathbf{I}_2$. The mechanism design for these two cases follows the approach established in the first case. ◇

Remark 1. According to (13) and the properties of the null space, one has that $\text{null}(M_a) = \text{null}(b_1^{\mathrm{T}})$ and $\text{null}(M_b) \supseteq \text{null}\left(\begin{bmatrix} b_2 & \cdots & b_\pi\end{bmatrix}^{\mathrm{T}}\right)$. This implies that the matrices N_a and N_b can be directly constructed using the basis vectors in (11).

Note that all dimensions of x must be contaminated to achieve the desired privacy requirements. The following theorem provides the necessary conditions to meet this privacy goal.

Theorem 1. *The designed noise contamination mechanism enables complete confidentiality for x (i.e., no dimension of x is exposed to the environment) during the computation of Mx, if the following conditions are satisfied:*

(1) The basis vector b_1 used to construct M_a in (13) is not a multiple of any standard basis vector in $\mathbb{R}^n$.

(2) The span of the basis vectors used to construct M_b in (13), i.e., $\text{span}\,(b_2, \ldots, b_\pi)$, does not include any standard basis vectors in $\mathbb{R}^n$.

Proof. The complete confidentiality for x requires that all dimensions of $N_a\mathbf{n}_a$ and $N_b\mathbf{n}_b$ are not always 0 across different noise realizations. Theorem 1 can be proved by contradiction. Suppose one dimension (the sth dimension) of $N_a\mathbf{n}_a$ is always 0, then the standard basis vector $\mathbf{e}_s$ is not in $\text{span}(N_a) = \text{null}(M_a) = \text{null}\left(b_1^{\mathrm{T}}\right) = \text{span}(b_1)^{\perp}$. This implies that $\mathbf{e}_s$ is in $\text{span}(b_1)$, meaning $b_1 = \lambda_2\mathbf{e}_s$ for some scalar λ_2. Thus, the first condition can be derived by contradiction. Similarly, suppose one dimension (sth dimension) of $N_b\mathbf{n}_b$ is always 0, then the standard basis vector $\mathbf{e}_s$ is not in $\text{span}(N_b) = \text{null}(M_b) \supseteq \text{null}\left(\begin{bmatrix} b_2 & \cdots & b_\pi\end{bmatrix}^{\mathrm{T}}\right) = \text{span}\,(b_2, \ldots, b_\pi)^{\perp}$. This implies that $\mathbf{e}_s$ is in $\text{span}\,(b_2, \ldots, b_\pi)$. Thus, the second condition can also be derived by contradiction. ◇

Example 2. The noise contamination mechanism is illustrated in this example. Consider the matrix $M = \begin{bmatrix} 1 & 1 \\ 0 & 1\end{bmatrix}$ and a non-zero vector $x \in \mathbb{R}^2$. For the first decomposition, the matrix M is decomposed as $M = \begin{bmatrix} 1 \\ 0\end{bmatrix} b_1^{\mathrm{T}} + \begin{bmatrix} 1 \\ 1\end{bmatrix} b_2^{\mathrm{T}}$ using the standard basis vectors $b_1 = \mathbf{e}_1$ and $b_2 = \mathbf{e}_2$. In this case, $\text{span}(N_a) = \text{null}(b_1^{\mathrm{T}}) = \text{span}(b_2)$ and $\text{span}(N_b) = \text{span}(b_1)$, so it follows that $N_a = \lambda_1 b_2$ and $N_b = \lambda_2 b_1$. It is observed that the first dimension of x is not privacy protected when using $x+N_a\mathbf{n}_a$, and the second dimension of x is also exposed to the environment when using $x + N_b\mathbf{n}_b$. For the second decomposition, the matrix M is decomposed as $M = \begin{bmatrix} 1 \\ 0.5\end{bmatrix} b_1^{\mathrm{T}} + \begin{bmatrix} 0 \\ -0.5\end{bmatrix} b_2^{\mathrm{T}}$ using the basis vectors $b_1 = \begin{bmatrix} 1 & 1\end{bmatrix}^{\mathrm{T}}$ and $b_1 = \begin{bmatrix} 1 & -1\end{bmatrix}^{\mathrm{T}}$, which satisfy the conditions in Theorem 1. In this case, $\text{span}(N_a) = \text{span}(b_2)$ and $\text{span}(N_b) = \text{span}(b_1)$, so it follows that $N_a = \lambda_1 b_2$ and $N_b = \lambda_2 b_1$. It is found that all dimensions of x are contaminated by the injected noises during the computation of Mx.

Remark 2. The example above demonstrates the design of a noise contamination mechanism that ensures complete confidentiality for a 2-dimensional matrix. In fact, complete confidentiality can be achieved for any matrix $M \in \mathbb{R}^{m\times n}$. This conclusion follows directly because all standard basis vectors in $\mathbb{R}^n$ can be replaced by other n basis vectors. For example, one can use basis vectors $b_1 = \begin{bmatrix} 1 & -1 & \cdots & -1\end{bmatrix}^{\mathrm{T}}$ and $b_i = \begin{bmatrix} 1 & 0 & \cdots & 1 & \cdots & 0\end{bmatrix}^{\mathrm{T}}$ $(i > 1)$ with only the first and the ith elements of b_i being 1.

Remark 3. Unlike the artificial noise injection mechanism in Goel and Negi (2008), which relies on a column-rank-deficient matrix, the proposed noise contamination mechanism is applicable to general matrix–vector multiplications. Furthermore, the complete confidentiality conditions in Theorem 1 can always be satisfied through appropriate matrix decomposition. This implies that our mechanism offers a higher degree of privacy protection, as no component of the private vector is exposed to the environment under these confidentiality conditions.

We are now prepared to address the problem of private weighted sum aggregation with centralized weights and hidden weights. The discussion focuses on the first case in Lemma 3 but can be extended to other cases. The following scheme is proposed for private weighted sum aggregation with centralized weights.

Lemma 4. *Given non-zero matrices $M_i \in \mathbb{R}^{m\times n}$ $(i \in \mathbb{N}_L)$ for an aggregator and non-zero vectors $x_i \in \mathbb{R}^n$ $(i = 1, 2, \ldots, L)$ for agents, the private weighted sum aggregation with centralized weights can be achieved using the following scheme:*

(1) The aggregator decomposes M_i as $M_i = M_{i,a} + M_{i,b}$ according to Theorem 1. Then, the aggregator searches for matrices $N_{i,a}$ and $N_{i,b}$ such that $\text{span}(N_{i,a}) = \text{null}(M_{i,a})$ and $\text{span}(N_{i,b}) = \text{null}(M_{i,b})$, and sends them to the ith agent.

(2) Each agent contaminates x_i to keep its privacy using $x_i+N_{i,a}\mathbf{n}_{i,a}$ and $x_i + N_{i,b}\mathbf{n}_{i,b}$, where $\mathbf{n}_{i,a}$ and $\mathbf{n}_{i,b}$ are randomly generated noises with appropriate dimensions. Then, the contaminated messages are sent to the aggregator.

(3) The aggregator computes the weighted sum aggregate by $\sum_{i=1}^{L}\left\{M_{i,a}(x_i + N_{i,a}\mathbf{n}_{i,a}) + M_{i,b}(x_i + N_{i,b}\mathbf{n}_{i,b})\right\}$.

Remark 4. The first step in Lemma 4 can be performed in advance as an offline setup step. Compared with the scheme in Agrawal et al. (2016) that uses the functional encryption for inner products, the proposed scheme based on the noise contamination mechanism is more computationally efficient. Unlike the secret

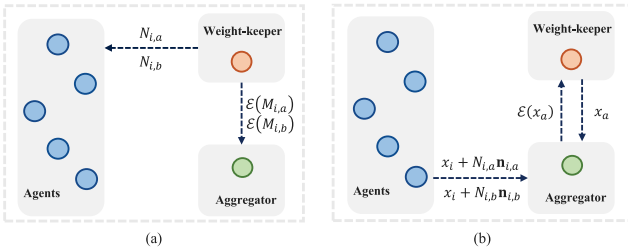

Fig. 1. (a) Offline setup for the scheme in Lemma 5; (b) Real-time aggregation for the scheme in Lemma 5.

sharing strategy, the proposed scheme does not require a dealer to distribute secret numbers and does not necessitate re-setup when agents are added or removed. Additionally, this scheme avoids the limitations associated with secret sharing for single agent.

Another scheme for private weighted sum aggregation with hidden weights is proposed in Lemma 5. This scheme includes an offline setup step and a real-time aggregation step, as illustrated in Fig. 1. The weight information in the aggregator is protected using the Paillier cryptosystem.

Lemma 5. *Given non-zero matrices $M_i \in \mathbb{R}^{m\times n}$ $(i \in \mathbb{N}_L)$ for a weight-keeper who does not participate in the aggregation and non-zero vectors $x_i \in \mathbb{R}^n$ $(i = 1, 2, \ldots, L)$ for agents, private weighted sum aggregation with hidden weights can be achieved by the following scheme:*

(1) The weight-keeper decomposes M_i as $M_i = M_{i,a} + M_{i,b}$ according to Theorem 1. To keep the weights secret, the weight-keeper encrypts $M_{i,a}$ and $M_{i,b}$ using the Paillier cryptosystem, resulting in $\mathcal{E}(M_{i,a})$ and $\mathcal{E}(M_{i,b})$, and transmits these encrypted matrices to the aggregator. Simultaneously, the weight-keeper searches for matrices $N_{i,a}$ and $N_{i,b}$ such that $\mathrm{span}(N_{i,a}) = \mathrm{null}(M_{i,a})$ and $\mathrm{span}(N_{i,b}) = \mathrm{null}(M_{i,b})$, and sends these matrices to the ith agent.

(2) Each agent contaminates x_i to keep its privacy using $x_i + N_{i,a}\mathbf{n}_{i,a}$ and $x_i + N_{i,b}\mathbf{n}_{i,b}$. These contaminated messages are then sent to the aggregator.

(3) The aggregator computes the weighted sum aggregate in ciphertext $\mathcal{E}(x_a)$ by

$$\begin{aligned}\mathcal{E}\left(\sum_{i=1}^{L} M_i x_i\right) &= \boxplus_{\substack{i\in\mathbb{N}_L\\ s\in\{a,b\}}} \mathcal{E}\left(M_{i,s}(x_i + N_{i,s}\mathbf{n}_{i,s})\right)\\ &= \boxplus_{\substack{i\in\mathbb{N}_L\\ s\in\{a,b\}}} \left(\mathcal{E}\left(M_{i,s}\right) \boxtimes (x_i + N_{i,s}\mathbf{n}_{i,s})\right).\end{aligned} \tag{16}$$

(4) Finally, the weight-keeper assists the aggregator in decrypting the ciphertext by computing $x_a = \mathcal{D}(\mathcal{E}(x_a))$.

Remark 5. The scheme proposed in Lemma 5 is a double-layer encryption scheme. Unlike the scheme in Alexandru and Pappas (2022), this paper utilizes private matrix–vector multiplication based on the noise contamination mechanism for outer-layer encryption. This design avoids the drawbacks of secret sharing and prevents the exposure of the private key used in inter-layer encryption to other entities, thereby enhancing the confidentiality of the weighting matrices. Additionally, the computational burden on the agents is reduced because they are not required to perform any encryption operations.

4.2. Privacy-preserving distributed estimation

In what follows, two privacy-preserving distributed estimators will be designed using the schemes in Lemmas 4–5.

4.2.1. Privacy-preserving distributed estimator i

The first estimator is based on private weighted sum aggregation with centralized weights. Consider the local estimator in (8)–(9), each subsystem is regarded as an aggregator, and its neighbors are treated as agents. In the offline setup step, subsystems decompose their system matrices as $A_{i,i_\rho}(k) = A^a_{i,i_\rho}(k) + A^b_{i,i_\rho}(k)$ using Lemma 3. Then, subsystems determine the null spaces of these decomposed matrices and transmit the results to their neighbors. In the running step, subsystems use the null spaces to contaminate their local estimates and transmit these contaminated estimates back to their out-neighbors. Specifically, the contaminated estimates from neighboring subsystems are denoted as $\hat{\mathcal{X}}^a_{i_\rho}(k) := \langle \hat{x}^a_{i_\rho}(k), R^a_{i_\rho}(k)\rangle$ and $\hat{\mathcal{X}}^b_{i_\rho}(k) := \langle \hat{x}^b_{i_\rho}(k), R^b_{i_\rho}(k)\rangle$ with

$$\begin{cases}\hat{x}^s_{i_\rho}(k) = \hat{x}_{i_\rho}(k) + N^s_{i,i_\rho}(k)\mathbf{n}^s_{i,i_\rho}(k)\\ R^s_{i_\rho}(k) = R_{i_\rho}(k) + N^s_{i,i_\rho}(k)\mathbf{m}^s_{i,i_\rho}(k).\end{cases} \tag{17}$$

Here, $s = a, b$ and $N^s_{i,i_\rho}(k)$ is from the null space of $A^s_{i,i_\rho}(k)$. The random noises $\mathbf{n}^s_{i,i_\rho}(k)$ and $\mathbf{m}^s_{i,i_\rho}(k)$ are generated with appropriate dimensions. Subsequently, subsystems can perform the prediction step in (8) as follows:

$$\begin{aligned}\hat{x}^p_i(k) =& A_i(k-1)\hat{x}_i(k-1)\\ &+ \sum_{i_\rho\in\Omega_i}\sum_{s\in\{a,b\}} A^s_{i,i_\rho}(k-1)\hat{x}^s_{i_\rho}(k-1)\\ R^p_i(k) =& \Big[\, A_i(k-1)R_i(k-1) \quad B_i(k-1)\\ &\sum_{s\in\{a,b\}} A^s_{i,i_1}(k-1)R^s_{i_1}(k-1) \quad \ldots\\ &\sum_{s\in\{a,b\}} A^s_{i,i_{\theta_i}}(k-1)R^s_{i_{\theta_i}}(k-1) \,\Big].\end{aligned} \tag{18}$$

Then, the computation procedures for the first privacy-preserving distributed estimator are summarized in Algorithm 1.

Algorithm 1 Privacy-Preserving Distributed Estimator I.

1: **Offline Setup**: Given $x_i(0) \in \hat{\mathcal{X}}_i(0)$ $(i = 1, 2, \ldots, L)$; Subsystems decompose system matrices by Lemma 3 to obtain $A^a_{i,i_\rho}(k)$ and $A^b_{i,i_\rho}(k)$; Subsystems construct $N^a_{i,i_\rho}(k)$ and $N^b_{i,i_\rho}(k)$ from the null spaces of the decomposed matrices and send them to neighbors;

2: **Prediction**: Subsystems contaminate their local estimates by (17) and send the contaminated estimates to out-neighbors; Subsystems collect the contaminated estimates $\hat{\mathcal{X}}^a_{i_\rho}(k)$ and $\hat{\mathcal{X}}^b_{i_\rho}(k)$ from neighbors, and implement the prediction step (18) to obtain $\mathcal{X}^p_i(k)$;

3: **Observation Update**: The observation update step in (9) is performed to compute $\hat{\mathcal{X}}_i(k)$;

4: Return to Step 2 and repeat Steps 2-3 to compute $\hat{\mathcal{X}}_i(k+1)$ $(i \in \mathbb{N}_L)$.

Remark 6. The first privacy-preserving distributed estimator in Algorithm 1 inherits the advantages of the private weighted sum aggregation scheme with centralized weights in Lemma 4. However, a potential privacy risk remains: subsystem $\mathbf{S}_{i_\rho}$ $(i_\rho \in \Omega_i)$ might still inadvertently reveal partial information from its computational intermediates directly to subsystem $\mathbf{S}_i$. For example, if subsystem $\mathbf{S}_i$ gains access to the computational intermediate $A_{i,i_\rho}(k)\hat{x}_{i_\rho}(k)$, it could potentially reconstruct partial or even complete information about $\hat{x}_{i_\rho}(k)$ by solving the relevant equations.

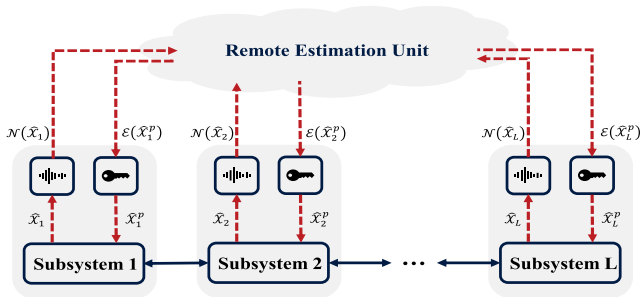

Fig. 2. Diagram of the second privacy-preserving distributed estimator for a cascading system.

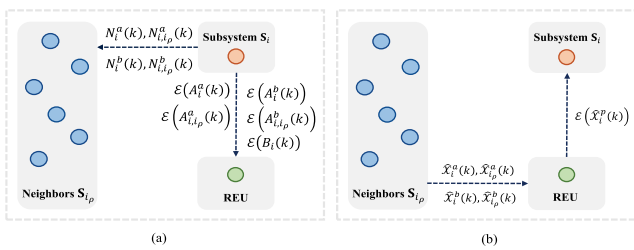

Fig. 3. (a) Offline setup for the privacy-preserving distributed estimator II; (b) Running step for the privacy-preserving distributed estimator II.

4.2.2. Privacy-preserving distributed estimator II

To enhance the privacy protection of the proposed distributed estimation algorithm, we then deploy the prediction step in (8) to a remote estimation unit (REU) managed by a third-party, as illustrated in Fig. 2 The introduction of a third-party is crucial because it addresses the privacy risks mentioned in Remark 6. Additionally, third-party computing services, such as cloud and fog computing (Darup et al., 2021; Darup & Jager, 2019), have been widely adopted in practical systems. As a consequence, more system designs are leveraging these efficient computing capabilities to reduce operational costs.

The third-party is assumed to be honest-but-curious, meaning it might retain private information during computation and communication for other purposes. Consequently, neither local estimates nor system matrices should be disclosed to the third-party. The privacy security of a subsystem is then defined as follows.

Definition 2. The subsystem $\mathbf{S}_i$ is privacy-protected if its private measurement $y_i(k)$ and private matrices $A_i(k)$, $A_{i,i_\rho}(k)$, $B_i(k)$, $C_i(k)$, and $D_i(k)$ are unavailable to other subsystems or untrusted entities.

As stated in Lemma 1, the problem of private weighted sum aggregation with centralized weights in a subsystem is transformed into the problem of private weighted sum aggregation with hidden weights in the third-party. Consequently, the scheme in Lemma 5 is applied to the privacy-preserving distributed estimation by treating subsystem $\mathbf{S}_i$ as a weight-keeper, the third-party as an aggregator, and neighboring subsystems as agents. The offline setup step and the running step for the privacy-preserving distributed estimator II are illustrated in Fig. 3.

In the offline setup step, subsystems decompose the system matrices according to Lemma 3 such that $A_i(k) = A_i^a(k) + A_i^b(k)$ and $A_{i,i_\rho}(k) = A_{i,i_\rho}^a(k) + A_{i,i_\rho}^b(k)$. They then encrypt and transmit these decomposed matrices, along with other system matrices, to the REU as $\mathcal{E}(A_i^a(k))$, $\mathcal{E}(A_i^b(k))$, $\mathcal{E}(A_{i,i_\rho}^a(k))$, $\mathcal{E}(A_{i,i_\rho}^b(k))$, and $\mathcal{E}(B_i(k))$. In this process, the information of subsystems is protected by the Paillier cryptosystem. Subsequently, subsystems construct $N_i^a(k)$, $N_i^b(k)$, $N_{i,i_\rho}^a(k)$, and $N_{i,i_\rho}^b(k)$ from the null spaces of the decomposed matrices and broadcast these matrices to neighboring subsystems.

In the running step, subsystems use the null spaces to contaminate their local estimates and transmit the contaminated estimates to the third-party. The contaminated estimates are denoted by $\hat{\mathcal{X}}_i^a(k) := \langle \hat{x}_i^a(k), R_i^a(k)\rangle$, $\hat{\mathcal{X}}_i^b(k) := \langle \hat{x}_i^b(k), R_i^b(k)\rangle$, $\hat{\mathcal{X}}_{i_\rho}^a(k)$, and $\hat{\mathcal{X}}_{i_\rho}^b(k)$ with representations similar to those in (17). The REU then performs the prediction step in ciphertext, where the encrypted prediction of the local state $\mathcal{E}\left(\hat{\mathcal{X}}_i^p(k)\right) := \langle \mathcal{E}(\hat{x}_i^p(k)), \mathcal{E}(R_i^p(k))\rangle$ is calculated as follows:

$$\begin{aligned}\mathcal{E}\left(\hat{x}_i^p(k)\right) &= \boxplus_{s\in\{a,b\}} \left(\mathcal{E}(A_i^s(k-1))\boxtimes\hat{x}_i^s(k-1)\right)\\ &\quad \boxplus_{\substack{i_\rho\in\Omega_i\\ s\in\{a,b\}}} \left(\mathcal{E}(A_{i,i_\rho}^s(k-1))\boxtimes\hat{x}_{i_\rho}^s(k-1)\right)\\ \mathcal{E}(R_i^p(k)) &= \Big[\boxplus_{s\in\{a,b\}} \left(\mathcal{E}(A_i^s(k-1))\boxtimes R_i^s(k-1)\right) \quad \mathcal{E}(B_i(k-1))\\ &\quad \boxplus_{s\in\{a,b\}} \left(\mathcal{E}(A_{i,i_1}^s(k-1))\boxtimes R_{i_1}^s(k-1)\right) \quad \cdots\\ &\quad \boxplus_{s\in\{a,b\}} \left(\mathcal{E}(A_{i,i_{\theta_i}}^s(k-1))\boxtimes R_{i_{\theta_i}}^s(k-1)\right)\Big].\end{aligned} \tag{19}$$

After the encrypted prediction step is completed, the REU transmits the computed prediction in ciphertext, $\mathcal{E}(\hat{\mathcal{X}}_i^p(k))$, back to subsystem $\mathbf{S}_i$. Subsequently, subsystem $\mathbf{S}_i$ decrypts the prediction in ciphertext, obtaining the prediction in plaintext, *i.e.*, $\hat{\mathcal{X}}_i^p(k) = \mathcal{D}(\mathcal{E}(\hat{\mathcal{X}}_i^p(k)))$. Once the measurement is collected from sensors, the observation update step in (9) is locally performed to compute $\hat{\mathcal{X}}_i(k)$. Then, subsystems inject noises into the obtained local zonotopic estimates using (17) and transmit these contaminated estimates to the REU as $\mathcal{N}\left(\hat{\mathcal{X}}_i(k)\right)$ for the computation of (19) at the next time instant.

The computation procedures for the second privacy-preserving distributed estimator are summarized in Algorithm 2.

Algorithm 2 Privacy-Preserving Distributed Estimator II.

1: **Offline Setup**: Given $x_i(0) \in \hat{\mathcal{X}}_i(0)$ $(i = 1, 2, \ldots, L)$; Subsystems encrypt the decomposed system matrices and send them to the REU as $\mathcal{E}(A_i^a(k))$, $\mathcal{E}(A_i^b(k))$, $\mathcal{E}(A_{i,i_\rho}^a(k))$, $\mathcal{E}(A_{i,i_\rho}^b(k))$, and $\mathcal{E}(B_i(k))$; Subsystems construct $N_i^a(k)$, $N_i^b(k)$, $N_{i,i_\rho}^a(k)$, and $N_{i,i_\rho}^b(k)$ from the null spaces of the decomposed system matrices and broadcast them to neighbors;

2: **Prediction**: The REU collects the contaminated local estimates $\hat{\mathcal{X}}_i^a(k)$, $\hat{\mathcal{X}}_i^b(k)$, $\hat{\mathcal{X}}_{i_\rho}^a(k)$, $\hat{\mathcal{X}}_{i_\rho}^b(k)$ from subsystems; The REU implements the encrypted prediction step in (19) to compute $\mathcal{E}(\hat{\mathcal{X}}_i^p(k))$ and then transmits the encrypted prediction $\mathcal{E}(\hat{\mathcal{X}}_i^p(k))$ back to subsystems;

3: **Observation Update**: Subsystem $\mathbf{S}_i$ receives the encrypted prediction and decrypts it into plaintext by $\hat{\mathcal{X}}_i^p(k) = \mathcal{D}(\mathcal{E}(\hat{\mathcal{X}}_i^p(k)))$; Subsystem $\mathbf{S}_i$ calculates the optimal gain matrix $K_i^{\text{opt}}(k)$ by (10) and collects the local measurement $y_i(k)$; The observation update step in plaintext is performed locally to obtain $\hat{\mathcal{X}}_i(k)$; Subsystem $\mathbf{S}_i$ injects noise signals into the local estimate as set $\mathcal{N}\left(\hat{\mathcal{X}}_i(k)\right)$ and transmits it to the REU;

4: Return to Step 2 and implement Steps 2-3 for calculating $\hat{\mathcal{X}}_i(k+1)$ $(i \in \mathbb{N}_L)$.

The computational complexity of Algorithm 1 is primarily due to the matrix inversion operation in (9), which has a complexity of $O(m_i^3)$. For Algorithm 2, the computational complexity is determined by the encryption and decryption operations, yielding a complexity of $O(p^3)$. The following theorem holds for the second privacy-preserving distributed estimator.

Theorem 2. *Based on the private weighted sum aggregation scheme in Lemma 5, the proposed privacy-preserving distributed estimator with the participation of a third-party in (19) ensures both the estimator and the subsystem are privacy-protected, as specified in Definitions 1–2.*

Proof. The proof of this theorem follows from Theorem 1. According to Theorem 1, the noise contamination mechanism ensures complete confidentiality of $\hat{\mathcal{X}}_i(k)$ during the computation of the encrypted prediction step. This means that no component of $\hat{x}_i(k)$ or $R_i(k)$ can be revealed from contaminated messages by the honest-but-curious third-party. Thus, the privacy of the estimator, as specified in Definition 1, is guaranteed. Additionally, the privacy of the subsystem, as specified in Definition 2, is ensured by the Paillier cryptosystem. ◇

Remark 7. In noise-based privacy protection mechanisms, the concept of signal-to-noise ratio (SNR) from communication theory (Goel & Negi, 2008) can also be used to evaluate privacy security. In our noise contamination mechanism, the injected noises can be sufficiently large to result in a low SNR, making it impossible to reconstruct the original signal from the contaminated one. Combined with Theorem 1, this implies that no element of $\hat{x}_i(k)$ or $R_i(k)$ can be reconstructed, therefore, remains unavailable to the third-party or other subsystems during the encrypted prediction step.

Remark 8. Although some fully homomorphic cryptosystems, such as Gentry's encryption scheme (Gentry, 2009), can be employed to design privacy-preserving distributed estimators, they inherently face significant computational complexity issues. However, this problem can be mitigated by using the Paillier cryptosystem. Additionally, the dimension of $R_i(k)$ tends to increase over time, which results in substantial costs for storage and computation. This issue can be addressed by employing zonotopic approximation techniques such as the box enclosure, the parallelotope enclosure (Vicino & Zappa, 1996), and the weighted version of box enclosure (Combastel, 2015).

5. Numerical examples

In this section, a numerical example is deployed to demonstrate the effectiveness of the proposed methods. Consider an interconnected system composed of 20 2-mass-2-spring mechanical subsystems (Battistelli, Chisci, & Gherardini, 2017) with randomly generated coupling terms. The state of the ith 2-mass-2-spring system is denoted by $x_i := \left[d_{i1}, \mathrm{v}_{i1}, d_{i2}, \mathrm{v}_{i2}\right]^{\mathrm{T}}$. Here, d_{i1} and d_{i2} are the displacements of the masses from their equilibrium positions, while v_{i1} and v_{i2} are the velocities of the masses. The dynamics of the ith mass–spring system can be described by the continuous-time linear state equation $\dot{x}_i(t) = A_{i,c}x_i(t)$, where $A_{i,c}$ is given by

$$A_{i,c} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{(k_{i1}+k_{i2})}{m_{i1}} & 0 & \frac{k_{i2}}{m_{i1}} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_{i2}}{m_{i2}} & 0 & -\frac{k_{i2}}{m_{i2}} & 0 \end{bmatrix}. \tag{20}$$

Here, m_{i1} and m_{i2} are the masses, while k_{i1} and k_{i2} are the stiffnesses of the springs. This continuous-time model can be discretized with a sampling period T, resulting in the discrete-time system $x_i(k+1) = A_i x_i(k)$ with $A_i = e^{A_{i,c}T}$. The coupling graph (Zhang et al., 2022) of this interconnected system is generated randomly to characterize the set Ω_i, and the coupling terms $A_{i,i_\rho}(k)$ ($i_\rho \in \Omega_i$) are also constructed randomly. Two sensors are deployed to observe the displacements d_{i1} and d_{i2} for each subsystem. It is assumed that the dynamics of the state and the measurement are disturbed by zonotopic noises, so this interconnected system is fully described by the model in (5).

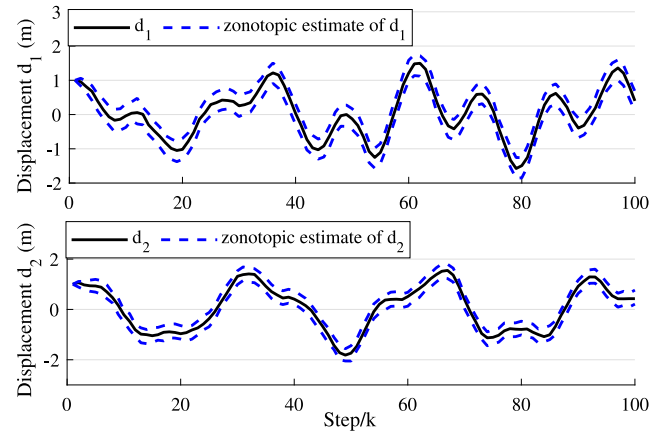

Fig. 4. The displacements and corresponding zonotopic estimates for the first subsystem.

In the simulation, the values of k_{i1} and k_{i2} are randomly extracted from the interval [5, 15], while m_{i1} and m_{i2} are uniformly distributed in the interval [0.5, 1.5]. The sampling period is set as $T = 0.1$, and the elements of the coupling matrices $A_{i,i_\rho}(k)$ ($i_\rho \in \Omega_i$) are selected from $[-0.001, 0.001]$. The noise terms $B_i(k)w_i(k)$ and $D_i(k)v_i(k)$ are wrapped within $\langle \mathbf{0}, 0.1\mathbf{I} \rangle$. The simulation is performed using Matlab R2020a on a 2.9 GHz Intel Core i5 processor. To invoke Python's libraries for operations involving large numbers, the environment of Python 3.7 (corresponding to the Matlab version) is also required. The Paillier cryptosystem is implemented using functions from the module "phe.paillier ", and the encrypted operations in (3) are programmed accordingly. Notice that the negative and floating point numbers can be handled in the Paillier cryptosystem through an encoding–decoding mechanism, with its quantization effects discussed in Darup et al. (2019).

The proposed algorithms have been implemented in the simulation. The noise contamination mechanism relies on the decomposition of $A_{i,c}$ and $A_{i,i_\rho}(k)$. According to Theorem 1, the basis vectors can be constructed as $b_1 = [1\ 1\ 0\ 0]^{\mathrm{T}}$, $b_2 = [1\ 0\ 1\ 0]^{\mathrm{T}}$, $b_3 = [1\ 0\ 0\ 1]^{\mathrm{T}}$, and $b_4 = [1\ -1\ -1\ -1]^{\mathrm{T}}$. Since $b = [b_1\ b_2\ b_3\ b_4]$ is nonsingular, the matrix $\mathcal{A}$ in (12) can be determined by $\mathcal{A} = (b^{-1}M^{\mathrm{T}})^{\mathrm{T}}$, then the decomposition is achieved using (13). To show the effectiveness of the distributed zonotopic estimator in (8)–(9), the displacements and their estimated intervals for the first subsystem are plotted in Fig. 4. The estimated interval is the projection of the zonotopic estimate on a certain dimension (Combastel, 2015). The results show that the estimated interval effectively tracks the real states and does not diverge over time. The running times for the offline setup and each step of the real-time estimation algorithms are recorded separately. To demonstrate the relationship between the connectivity degree of a subsystem (*i.e.*, the cardinality of the set Ω_i ($i \in \Omega_i$)) and the computing time, we generate different coupling graphs to create various realizations of the interconnected system. The offline setup times for null space construction and weight encryption under different average connectivity degrees (*i.e.*, the average cardinality of sets Ω_i ($i = 1, 2, \ldots, L$)) are plotted in Fig. 5. Notice that Algorithm 1 only requires the setup for null space construction, whereas Algorithm 2 requires both steps. Fig. 6 illustrates the average, maximum, and minimum running times across all subsystems for executing Algorithm 2 under different average connectivity degrees. As the

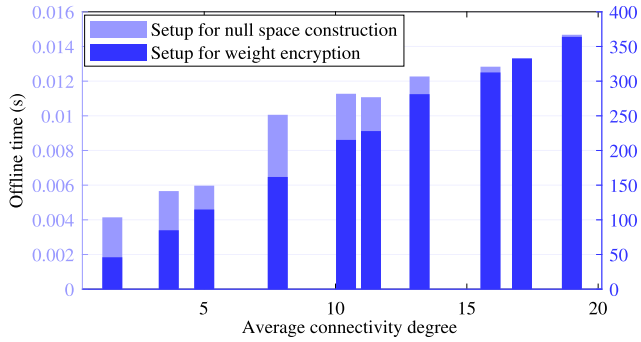

Fig. 5. Offline setup time for null space construction and weight encryption.

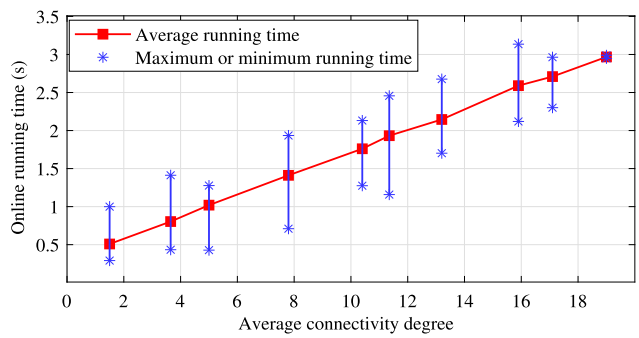

Fig. 6. The average, maximum, and minimum running time of Algorithm 6.2 across all subsystems.

number of subsystem couplings increases, the average running time for the distributed estimator also increases nearly linearly. The noise contamination mechanism is significantly more computationally efficient compared to traditional encryption algorithms, highlighting the superiority of Algorithm 1. For Algorithm 2, it achieves the privacy security defined in Definitions 1–2, albeit with more computing power.

6. Conclusion

This paper addressed the critical issue of privacy protection in distributed estimation for IDSs. We proposed a noise contamination mechanism for private matrix–vector multiplication and derived conditions to ensure complete confidentiality during the multiplication process. These conditions offer valuable guidance for generating artificial noises to achieve the desired level of privacy. Additionally, we introduced a scheme for private weighted sum aggregation with centralized weights by private matrix–vector multiplication, and designed a double-layer encryption scheme for private weighted sum aggregation with hidden weights by combining private matrix–vector multiplication and the Paillier cryptosystem. These schemes were applied to develop privacy-preserving distributed estimators. It is demonstrated that privacy security can be guaranteed for subsystems and their distributed estimators with third-party outsourcing. An illustrative example is provided to validate the effectiveness of the proposed noise contamination mechanism and privacy-preserving distributed estimators. Future work could explore the application of the proposed noise contamination mechanism in other privacy-sensitive scenarios, such as distributed model predictive control and distributed optimization.

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References

Agrawal, S., Libert, B., & Stehlé, D. (2016). Fully secure functional encryption for inner products, from standard assumptions. In *Advances in cryptology – CRYPTO 2016* (pp. 333–362). Santa Barbara, CA, USA: Springer Berlin Heidelberg.

Alexandru, A. B., & Pappas, G. J. (2022). Private weighted sum aggregation. *IEEE Transactions on Control of Network Systems*, *9*(1), 219–230.

Alexandru, A. B., Schulze Darup, M., & Pappas, G. J. (2019). Encrypted cooperative control revisited. In *2019 IEEE 58th conference on decision and control* (pp. 7196–7202). Nice, France: IEEE.

Battistelli, G., Chisci, L., & Gherardini, S. (2017). Moving horizon estimation for discrete-time linear systems with binary sensors: Algorithms and stability results. *Automatica*, *85*, 374–385.

Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H. B., Patel, S., et al. (2017). Practical secure aggregation for privacy-preserving machine learning. In *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security* (pp. 1175–1191). Dallas Texas USA: ACM.

Chen, B., Hu, G., Ho, D. W. C., & Yu, L. (2019). Distributed Kalman filtering for time-varying discrete sequential systems. *Automatica*, *99*, 228–236.

Chen, B., Hu, G., Ho, D. W., & Yu, L. (2022). Distributed estimation and control for discrete time-varying interconnected systems. *IEEE Transactions on Automatic Control*, *67*(5), 2192–2207.

Combastel, C. (2015). Zonotopes and Kalman observers: Gain optimality under distinct uncertainty paradigms and robust convergence. *Automatica*, *55*, 265–273.

Cortés, J., Dullerud, G. E., Han, S., Le Ny, J., Mitra, S., & Pappas, G. J. (2016). Differential privacy in control and network systems. In *2016 IEEE 55th conference on decision and control (CDC)* (pp. 4252–4272). Las Vegas, USA: IEEE.

Creaco, E., Campisano, A., Fontana, N., Marini, G., Page, P., & Walski, T. (2019). Real time control of water distribution networks: A state-of-the-art review. *Water Research*, *161*, 517–530.

Darup, M. S., Alexandru, A. B., Quevedo, D. E., & Pappas, G. J. (2021). Encrypted control for networked systems: An illustrative introduction and current challenges. *IEEE Control Systems Magazine*, *41*(3), 58–78.

Darup, M. S., & Jager, T. (2019). Encrypted cloud-based control using secret sharing with one-time pads. In *2019 IEEE 58th conference on decision and control* (pp. 7215–7221). Nice, France: IEEE.

Darup, M. S., Redder, A., & Quevedo, D. E. (2019). Encrypted cooperative control based on structured feedback. *IEEE Control Systems Letters*, *3*(1), 37–42.

Farina, M., Ferrari-Trecate, G., & Scattolini, R. (2010). Moving-horizon partition-based state estimation of large-scale systems. *Automatica*, *46*(5), 910–918.

Farokhi, F., Shames, I., & Batterham, N. (2017). Secure and private control using semi-homomorphic encryption. *Control Engineering Practice*, *67*, 13–20.

Gentry, C. (2009). Fully homomorphic encryption using ideal lattices. In *Proceedings of the forty-first annual ACM symposium on theory of computing* (pp. 169–178). Bethesda MD USA: ACM.

Giaconi, G., Gunduz, D., & Poor, H. V. (2018). Privacy-aware smart metering: Progress and challenges. *IEEE Signal Processing Magazine*, *35*(6), 59–78.

Goel, S., & Negi, R. (2008). Guaranteeing secrecy using artificial noise. *IEEE Transactions on Wireless Communications*, *7*(6), 2180–2189.

Haber, A., & Verhaegen, M. (2013). Moving horizon estimation for large-scale interconnected systems. *IEEE Transactions on Automatic Control*, *58*(11), 2834–2847.

Joye, M., & Libert, B. (2013). A scalable scheme for privacy-preserving aggregation of time-series data. In *Financial cryptography and data security* (pp. 111–125). Okinawa, Japan: Springer Berlin Heidelberg.

Jung, D. H., Ryu, J. G., & Choi, J. (2022). When satellites work as eavesdroppers. *IEEE Transactions on Information Forensics and Security*, *17*, 2784–2799.

Kekatos, V., & Giannakis, G. B. (2013). Distributed robust power system state estimation. *IEEE Transactions on Power Systems*, *28*(2), 1617–1626.

Khan, U. A., & Moura, J. M. F. (2008). Distributing the Kalman filter for large-scale systems. *IEEE Transactions on Signal Processing*, *56*(10), 4919–4935.

Li, S. E., Zheng, Y., Li, K., Wu, Y., Hedrick, J. K., Gao, F., et al. (2017). Dynamical modeling and distributed control of connected and automated vehicles: Challenges and opportunities. *IEEE Intelligent Transportation Systems Magazine*, *9*(3), 46–58.

Lian, J., Yang, T., & Yuan, Y. (2019). Special section on control of complex networked systems (CCNS): Recent results and future trends. *Annual Reviews in Control*, *47*, 275–277.

Mo, Y., & Murray, R. M. (2017). Privacy preserving average consensus. *IEEE Transactions on Automatic Control*, *62*(2), 753–765.

Nozari, E., Tallapragada, P., & Cortés, J. (2015). Differentially private average consensus with optimal noise selection. *IFAC-PapersOnLine*, *48*(22), 203–208.

Nozari, E., Tallapragada, P., & Cortés, J. (2017). Differentially private average consensus: Obstructions, trade-offs, and optimal algorithm design. *Automatica*, *81*, 221–231.

Ny, J. L., & Pappas, G. J. (2014). Differentially private filtering. *IEEE Transactions on Automatic Control*, *59*(2), 341–354.

Riverso, S., Farina, M., Scattolini, R., & Ferrari-Trecate, G. (2013). Plug-and-play distributed state estimation for linear systems. In *52nd IEEE conference on decision and control* (pp. 4889–4894). Florence, Italy: IEEE.

Riverso, S., Rubini, D., & Ferrari-Trecate, G. (2015). Distributed bounded-error state estimation based on practical robust positive invariance. *International Journal of Control*, *88*(11), 2277–2290.

Ruan, M., Gao, H., & Wang, Y. (2019). Secure and privacy-preserving consensus. *IEEE Transactions on Automatic Control*, *64*(10), 4035–4049.

Shamir, A. (1979). How to share a secret. *Communications of the ACM*, *22*(11), 612–613.

Shim, K. A. (2016). A survey of public-key cryptographic primitives in wireless sensor networks. *IEEE Communications Surveys & Tutorials*, *18*(1), 577–601.

Sultangazin, A., & Tabuada, P. (2021). Symmetries and isomorphisms for privacy in control over the cloud. *IEEE Transactions on Automatic Control*, *66*(2), 538–549.

Tjell, K., & Wisniewski, R. (2020). Privacy preserving distributed summation in a connected graph. *IFAC-PapersOnLine*, *53*(2), 3445–3450.

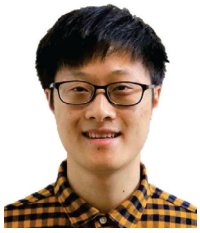
Tsiamis, A., Gatsis, K., & Pappas, G. J. (2020). State-secrecy codes for networked linear systems. *IEEE Transactions on Automatic Control*, *65*(5), 2001–2015.

Vicino, A., & Zappa, G. (1996). Sequential approximation of feasible parameter sets for identification with set membership uncertainty. *IEEE Transactions on Automatic Control*, *41*(6), 774–785.

Yan, X., Zhang, Y., Xu, D., & Chen, B. (2022). Distributed confidentiality fusion estimation against eavesdroppers. *IEEE Transactions on Aerospace and Electronic Systems*, *58*(4), 3633–3642.

Zhang, Y., Chen, B., & Yu, L. (2023). Distributed zonotopic estimation for interconnected systems: A fusing overlapping states strategy. *Automatica*, *155*, Article 111144.

Zhang, Y., Chen, B., Yu, L., & Ho, D. W. C. (2022). Distributed Kalman filtering for interconnected dynamic systems. *IEEE Transactions on Cybernetics*, *52*(11), 11571–11580.



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